# Thadomal Shahani Engineering College

## Departement of Artificial Intelligence and Data Science

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## EXPERIMENT 7: To Implement monitoring using Prometheus / Grafana / Nagios.

**Aim:** To implement monitoring using Prometheus / Grafana / Nagios
**LO Mapping:** LO5
**LO5**: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

**Theory:**

Continuous monitoring involves continuously checking an organization's IT infrastructure to identify, report, and respond to issues or potential security threats. After an application is deployed on a server, continuous monitoring helps ensure that the system remains secure, stable, and available.

**Why Do We Need the Nagios Tool?**

Nagios is a popular, powerful, open-source monitoring tool used to monitor servers, networks, applications, and other IT infrastructure components. Its main benefits include:

- Detects network failures and server-related issues.
- Helps identify the root cause of problems and find permanent solutions.
- Monitors the entire IT infrastructure and business processes.
- Identifies server performance issues and supports troubleshooting.
- Helps plan infrastructure upgrades before outdated systems cause failures.
- Improves service availability and supports security monitoring.
- Can execute automated recovery actions when configured to do so.
- Helps organizations detect and resolve IT problems before they affect critical business operations.

**NRPE and NSClient++**

NRPE stands for Nagios Remote Plugin Executor. It allows Nagios to execute monitoring plugins on remote systems and collect information about their performance.

NSClient++ is a monitoring agent used on Windows machines. It can check local resources such as CPU usage, memory, disk space, and running services, and send the results to the Nagios server.

NSClient++ runs as a Windows service and waits for monitoring requests from Nagios. When the server requests a check, the agent executes it locally and returns the results to Nagios for analysis.

**Note on Running Nagios on Windows**

Nagios Core is primarily designed for Linux and does not run natively on Windows. To use it on a Windows system, it can be run inside a Docker container using Docker Desktop with the WSL2 backend. Windows machines can then be monitored using NSClient++, which supports communication with Nagios through compatible protocols such as NRPE.

The commands used for setup and configuration can be executed through Windows PowerShell or Command Prompt, depending on the installation steps.

## 1 - Pre-requisite: Install Docker Desktop

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:

PS C:\Users\vishal> docker --version

PS C:\Users\vishal> docker version

Also make sure the WSL2 feature is enabled:

PS C:\Users\vishal> wsl --install

PS C:\Users\vishal> wsl --set-default-version 2

## 2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

PS C:\Users\vishal> docker pull jasonrivers/nagios:latest

PS C:\Users\vishal> docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest

Verify the container is up and running:

PS C:\Users\vishal> docker ps

## 3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:

PS C:\Users\vishal> docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin

Enter and confirm the password when prompted (e.g. 123456).

## 4 - Access Nagios Web Interface

Access the Nagios web interface from a browser on the Windows host:

http://localhost:8080/nagios/

Prompting for Authentication Password -

username**:** nagiosadmin

Password**:** 123456 (which you entered while configuration)

## 5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

PS C:\Users\vishal> Invoke-WebRequest -Uri "https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi" -OutFile "NSCP.msi"

PS C:\Users\vishal> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait

## 6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

PS C:\Program Files\NSClient++> notepad nsclient.ini allowed
hosts=127.0.0.1,192.168.1.100

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

After saving the file, restart the NSClient++ service so the change takes effect:

PS C:\Users\vishal> Restart-Service nscp

## 7 - Verify Connectivity from Nagios

From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

PS C:\Users\vishal> docker exec -it nagios4
/opt/nagios/libexec/check_nrpe -H
192.168.1.50 NRPE v2.15

## 8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

PS C:\Users\vishal> docker exec -it nagios4 sh -c "vi
/opt/nagios/etc/servers/MyWindowsHost001.cfg"

```
define host {
   use             windows-server     host_name
Windows_Host_001     alias           Windows
Host 001     address          192.168.1.50
   register         1
}
define service{
   host_name          Windows_Host_001    service_description    PING
check_command                        check_ping!100.0,20%!500.0,60%
max_check_attempts      2       check_interval          2       retry_interval
2    check_period        24x7    check_freshness      1    contact_groups
admins     notification_interval  2     notification_period    24x7
   notifications_enabled  1
   register              1
}
```

Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

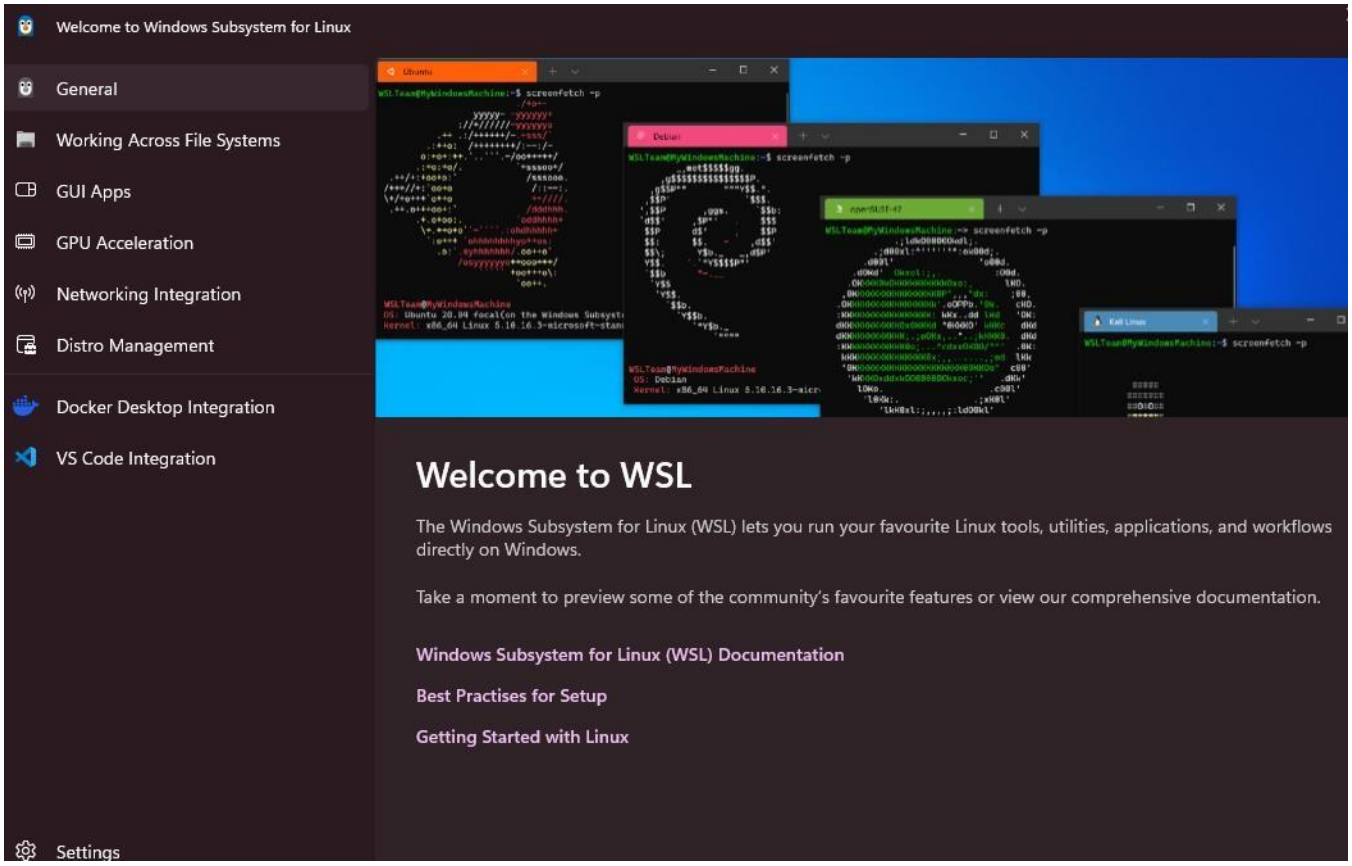
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg

PS C:\Users\vishal> docker restart nagios4

## 9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

```
PS C:\Users\HP 15L> wsl --install -d Ubuntu
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: hp_15l
```



```
Create a default UNIX user account: hp_15l
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!

Help us improve Ubuntu features and compatibility by sharing system reports with Canonical.
Reports are sent anonymously and do not contain any personal data.
For legal details, please visit: https://ubuntu.com/legal/systems-information-notice

We will save your answer to Windows and will only ask you once.

Would you like to opt-in to platform metrics collection (Y/n)? To see an example of the data collected, enter
'e'.
[Y/n/e]: y
```

```
[Y/n/e]: y
hp_15l@SExYSagar:/mnt/c/Users/HP 15L$ sudo apt update && sudo apt install -y autoconf gcc libc6 make wget unzip apache2 php libapache2-mod-php libgd-dev openssl libssl-dev
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [501 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [117 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [54.8 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [418 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [81.5 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [10.8 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2844 B]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:18 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:19 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [290 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:26 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [640 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [153 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [295 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [94.0 kB]
Get:31 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:33 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [431 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [84.5 kB]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3172 B]
```

```
88 packages can be upgraded. Run 'apt list --upgradable' to see them.
wget is already the newest version (1.25.0-2ubuntu4.4).
wget set to manually installed.
Upgrading:
  bsdextrautils  libc-gconv-modules-extra  libmount1      openssl
  bsdutils       libc6                     libsmartcols1  openssl-provider-legacy
  eject          libexpat1                 libssl3t64     util-linux
  fdisk          libfdisk1                 libuuid1       zlib1g
  libblkid1      libheif-plugin-aomdec     locales
  libbz2-1.0     libheif-plugin-aomenc     login
  libc-bin       libheif1                  mount

Installing:
  apache2   gcc                 libgd-dev   make  unzip
  autoconf  libapache2-mod-php  libssl-dev  php

Installing dependencies:
  apache2-bin              libc6-dev           liblerc4         libxcb1-dev
  apache2-data             libcc1-0            liblsan0         libxdmcp-dev
  apache2-utils            libdeflate-dev      liblua5.4-0      libxpm-dev
  automake                 libdeflate0         liblzma-dev      libxpm4
  autotools-dev            libexpat1-dev       libmpc3          libzstd-dev
  bzip2-doc                libfontconfig-dev   libpkgconf7      linux-libc-dev
  cpp                      libfreetype-dev     libpng-dev       m4
  cpp-15                   libgcc-15-dev       libpng-tools     manpages-dev
  cpp-15-x86-64-linux-gnu  libgd3              libquadmath0     php-common
  cpp-x86-64-linux-gnu     libgomp1            libsharpyuv-dev  php8.5
  gcc-15                   libheif-dev         libtiff-dev      php8.5-cli
  gcc-15-base              libhwasan0          libtiff6         php8.5-common
  gcc-15-x86-64-linux-gnu  libimagequant-dev   libtiffxx6       php8.5-readline
  gcc-x86-64-linux-gnu     libimagequant0      libtsan2         pkgconf
  libapache2-mod-php8.5    libisl23            libubsan1        pkgconf-bin
  libapr1t64               libitm1             libvpx-dev       rpcsvc-proto
  libaprutil1-dbd-sqlite3  libjbig-dev         libvpx12         ssl-cert
  libaprutil1-ldap         libjbig0            libwebp-dev      uuid-dev
  libaprutil1t64           libjpeg-dev         libwebp7         x11proto-dev
  libargon2-1              libjpeg-turbo8      libwebpdecoder3  xorg-sgml-doctools
  libasan8                 libjpeg-turbo8-dev  libwebpdemux2    xtrans-dev
  libbrotli-dev            libjpeg8            libwebpmux3      zlib1g-dev
  libbz2-dev               libjpeg8-dev        libx11-dev
  libc-dev-bin             liblerc-dev         libxau-dev

Suggested packages:
  apache2-doc              gettext         gcc-15-multilib       libssl-doc
```

```
Suggested packages:
  apache2-doc              gettext         gcc-15-multilib       libssl-doc
  apache2-suexec-pristine  cpp-doc         gcc-15-doc            libx11-doc
  | apache2-suexec-custom  gcc-15-locales  gdb-x86-64-linux-gnu  libxcb-doc
  ufw                      cpp-15-doc      php-pear              m4-doc
  www-browser              gcc-multilib    libc-devtools         make-doc
  autoconf-archive         flex            glibc-doc             zip
  gnu-standards            bison           freetype2-doc
  autoconf-doc             gdb             libgd-tools
  libtool                  gcc-doc         liblzma-doc

Summary:
  Upgrading: 25, Installing: 103, Removing: 0, Not Upgrading: 63
  Download size: 107 MB
  Space needed: 325 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-gconv-modules-extra amd64 2.43-2ubuntu2.4 [1362 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6 amd64 2.43-2ubuntu2.4 [2104 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 eject amd64 2.41.3-3ubuntu2.2 [40.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfdisk1 amd64 2.41.3-3ubuntu2.2 [164 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bsdextrautils amd64 2.41.3-3ubuntu2.2 [86.7 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsmartcols1 amd64 2.41.3-3ubuntu2.2 [84.9 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 fdisk amd64 2.41.3-3ubuntu2.2 [139 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libblkid1 amd64 2.41.3-3ubuntu2.2 [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libmount1 amd64 2.41.3-3ubuntu2.2 [159 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mount amd64 2.41.3-3ubuntu2.2 [136 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libuuid1 amd64 2.41.3-3ubuntu2.2 [36.7 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 util-linux amd64 2.41.3-3ubuntu2.2 [1070 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-bin amd64 2.43-2ubuntu2.4 [701 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 login amd64 1:4.16.0-2+really2.41.3-3ubuntu2.2 [85.9 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libexpat1 amd64 2.7.4-1ubuntu0.1 [96.0 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libbz2-1.0 amd64 1.0.8-6ubuntu0.1 [36.2 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssl-provider-legacy amd64 3.5.5-1ubuntu3.5 [39.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 zlib1g amd64 1:1.3.dfsg+really1.3.1-1ubuntu3.1 [61.6 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libssl3t64 amd64 3.5.5-1ubuntu3.5 [2363 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bsdutils amd64 1:2.41.3-3ubuntu2.2 [80.7 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 locales all 2.43-2ubuntu2.4 [4225 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssl amd64 3.5.5-1ubuntu3.5 [1243 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/main amd64 libapr1t64 amd64 1.7.6-3 [109 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1t64 amd64 1.6.3-3ubuntu3.1 [93.3 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3.1 [11.4 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1-ldap amd64 1.6.3-3ubuntu3.1 [9228 B]
Get:27 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblua5.4-0 amd64 5.4.8-1build1 [163 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-bin amd64 2.4.66-2ubuntu2.4 [1355 kB]
```

```
Get:127 http://archive.ubuntu.com/ubuntu resolute/main amd64 ssl-cert all 1.1.3ubuntu2 [18.8 kB]
Get:128 http://archive.ubuntu.com/ubuntu resolute/main amd64 unzip amd64 6.0-29ubuntu1 [180 kB]
Fetched 107 MB in 36s (2938 kB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../libc6_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc6:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../eject_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking eject (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Preparing to unpack .../libfdisk1_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking libfdisk1:amd64 (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Preparing to unpack .../bsdextrautils_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking bsdextrautils (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Preparing to unpack .../libsmartcols1_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking libsmartcols1:amd64 (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Setting up libsmartcols1:amd64 (2.41.3-3ubuntu2.2) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../fdisk_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking fdisk (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Preparing to unpack .../libblkid1_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking libblkid1:amd64 (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Setting up libblkid1:amd64 (2.41.3-3ubuntu2.2) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../libmount1_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking libmount1:amd64 (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Setting up libmount1:amd64 (2.41.3-3ubuntu2.2) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../mount_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking mount (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Preparing to unpack .../libuuid1_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking libuuid1:amd64 (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Setting up libuuid1:amd64 (2.41.3-3ubuntu2.2) ...
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../util-linux_2.41.3-3ubuntu2.2_amd64.deb ...
Unpacking util-linux (2.41.3-3ubuntu2.2) over (2.41.3-3ubuntu2) ...
Setting up util-linux (2.41.3-3ubuntu2.2) ...
```

```
Setting up uuid-dev:amd64 (2.41.3-3ubuntu2.2) ...
Setting up zlib1g-dev:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.1) ...
Setting up cpp (4:15.2.0-5ubuntu1) ...
Setting up php8.5-cli (8.5.4-0ubuntu1.3) ...
update-alternatives: using /usr/bin/php8.5 to provide /usr/bin/php (php) in auto mode
update-alternatives: using /usr/bin/phar8.5 to provide /usr/bin/phar (phar) in auto mode
update-alternatives: using /usr/bin/phar.phar8.5 to provide /usr/bin/phar.phar (phar.phar) in auto mode
Creating config file /etc/php/8.5/cli/php.ini with new version
Setting up libjpeg8-dev:amd64 (8c-2ubuntu12) ...
Setting up apache2 (2.4.66-2ubuntu2.4) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/systemd/system/apache-htcacheclean.service'.
Setting up libpng-dev:amd64 (1.6.57-1) ...
Setting up libjpeg-dev:amd64 (8c-2ubuntu12) ...
Setting up gcc (4:15.2.0-5ubuntu1) ...
Setting up libfreetype-dev:amd64 (2.14.2+dfsg-1ubuntu0.1) ...
Setting up libtiff-dev:amd64 (4.7.0-3ubuntu5) ...
Setting up libapache2-mod-php8.5 (8.5.4-0ubuntu1.3) ...
Creating config file /etc/php/8.5/apache2/php.ini with new version
```

```
Setting up libapache2-mod-php8.5 (8.5.4-0ubuntu1.3) ...
Creating config file /etc/php/8.5/apache2/php.ini with new version
Module mpm_event disabled.
Enabling module mpm_prefork.
apache2_switch_mpm Switch to prefork
apache2_invoke: Enable module php8.5
Setting up php8.5 (8.5.4-0ubuntu1.3) ...
Setting up libapache2-mod-php (2:8.5+99ubuntu1) ...
Setting up php (2:8.5+99ubuntu1) ...
Setting up libfontconfig-dev:amd64 (2.17.1-3ubuntu1) ...
Setting up libheif-plugin-aomdec:amd64 (1.21.2-3ubuntu0.5) ...
Setting up libheif1:amd64 (1.21.2-3ubuntu0.5) ...
Setting up libheif-dev:amd64 (1.21.2-3ubuntu0.5) ...
Setting up libgd3:amd64 (2.3.3-13ubuntu2) ...
Setting up libheif-plugin-aomenc:amd64 (1.21.2-3ubuntu0.5) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for sgml-base (1.31+nmu1build1) ...
Processing triggers for install-info (7.2-5ubuntu2) ...
Setting up x11proto-dev (2025.1-1) ...
Setting up libxau-dev:amd64 (1:1.0.11-1build2) ...
Setting up libxdmcp-dev:amd64 (1:1.1.5-2) ...
Setting up libxcb1-dev:amd64 (1.17.0-2ubuntu1) ...
Setting up libx11-dev:amd64 (2:1.8.13-1) ...
Setting up libxpm-dev:amd64 (1:3.5.17-1ubuntu0.26.04.1) ...
Setting up libgd-dev:amd64 (2.3.3-13ubuntu2) ...
Processing triggers for php8.5-cli (8.5.4-0ubuntu1.3) ...
Processing triggers for libapache2-mod-php8.5 (8.5.4-0ubuntu1.3) ...
hp_15l@SExYSagar:/mnt/c/Users/HP 15L$
```

```
../common/downtime.c:439:17: warning: ignoring return value of ‘asprintf’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
  439 |                 asprintf(&temp_buffer, "This %s has been scheduled for fixed downtime from %s to %s.  Notifications for the %s will not be sent out during that time p
eriod.", type_string, start_time_string, end_time_string, type_string);
      |                 ^~~~
```

```
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/css/bootstrap-theme.min.css
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/css/bootstrap.css
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/css/bootstrap.css.map
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/css/bootstrap.min.css
   creating: angular-1.3.9/docs/components/bootstrap-3.1.1/fonts/
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/fonts/glyphicons-halflings-regular.eot
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/fonts/glyphicons-halflings-regular.svg
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/fonts/glyphicons-halflings-regular.ttf
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/fonts/glyphicons-halflings-regular.woff
   creating: angular-1.3.9/docs/components/bootstrap-3.1.1/js/
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/js/bootstrap.js
  inflating: angular-1.3.9/docs/components/bootstrap-3.1.1/js/bootstrap.min.js
   creating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/CHANGES.html
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/COPYING
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/README-zh-Hans.html
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/README.html
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/README.md
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/bower.json
   creating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-apollo.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-clj.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-css.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-dart.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-go.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-hs.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-lisp.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-lua.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-ml.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-n.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-proto.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-scala.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-sql.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-tex.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-vb.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-vhdl.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-wiki.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-xq.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/lang-yaml.js
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/prettify.css
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/src/prettify.js
   creating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/styles/
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/styles/demo.html
  inflating: angular-1.3.9/docs/components/google-code-prettify-1.0.1/styles/desert.css
```

```
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/module'
cd ./worker && make all
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.14/worker'
cd ./ping && make all
make[2]: Entering directory '/tmp/nagioscore-nagios-4.4.14/worker/ping'
gcc -I.. -I ../../include -I ../.. -g -O2  -DHAVE_CONFIG_H -o worker-ping worker-ping.c -L ../../lib -l nagios  -lssl -lcrypto
worker-ping.c: In function ‘worker’:
worker-ping.c:238:17: warning: ignoring return value of ‘read’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
  238 |                 read( sd, response + 3, sizeof(response) - 4);
      |                 ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
worker-ping.c: In function ‘daemon_init’:
worker-ping.c:361:17: warning: ignoring return value of ‘chdir’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
  361 |                 chdir( "/");
      |                 ^~~~~~~~~~
make[2]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/worker/ping'
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/worker'

*** Compile finished ***

If the main program and CGIs compiled without any errors, you
can continue with testing or installing Nagios as follows (type
'make' without any arguments for a list of all possible options):

  make test
     - This runs the test suite

  make install
     - This installs the main program, CGIs, and HTML files

  make install-init
     - This installs the init script in /lib/systemd/system

  make install-daemoninit
     - This will initialize the init script
       in /lib/systemd/system

  make install-groups-users
     - This adds the users and groups if they do not exist

  make install-commandmode
     - This installs and configures permissions on the
       directory for holding the external command file

  make install-config
```

```
  make install-exfoliation
     - This installs the Exfoliation theme for the Nagios
       web interface

  make install-classicui
     - This installs the classic theme for the Nagios
       web interface


*** Support Notes *****************************************

If you have questions about configuring or running Nagios,
please make sure that you:

     - Look at the sample config files
     - Read the documentation on the Nagios Library at:
          https://library.nagios.com

before you post a question to one of the mailing lists.
Also make sure to include pertinent information that could
help others help you.  This might include:

     - What version of Nagios you are using
     - What version of the plugins you are using
     - Relevant snippets from your config files
     - Relevant error messages from the Nagios log file

For more information on obtaining support for Nagios, visit:

       https://support.nagios.com

************************************************************

Enjoy.

hp_15l@SExYSagar:/tmp/nagioscore-nagios-4.4.14$
```

```
hp_15l@SExYSagar:/tmp/nagioscore-nagios-4.4.14$ sudo make install-groups-users
sudo usermod -a -G nagios www-data
sudo make install
sudo make install-daemoninit
sudo make install-commandmode
sudo make install-config
sudo make install-webconf
groupadd -r nagios
useradd -g nagios nagios
cd ./base && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.14/base'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagios /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagiostats /usr/local/nagios/bin
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/base'
cd ./cgi && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.14/cgi'
make install-basic
make[2]: Entering directory '/tmp/nagioscore-nagios-4.4.14/cgi'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/sbin
for file in *.cgi; do \
        /usr/bin/install -c -s -m 775 -o nagios -g nagios $file /usr/local/nagios/sbin; \
done
make[2]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/cgi'
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.14/cgi'
cd ./html && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.14/html'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/media
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/stylesheets
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/contexthelp
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/js
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images/logos
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/includes
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/ssi
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs/angular-1.3.9
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs/ui-utils-0.2.3
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/bootstrap-3.3.7
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/bootstrap-3.3.7/css
```

```
*** Config files installed ***

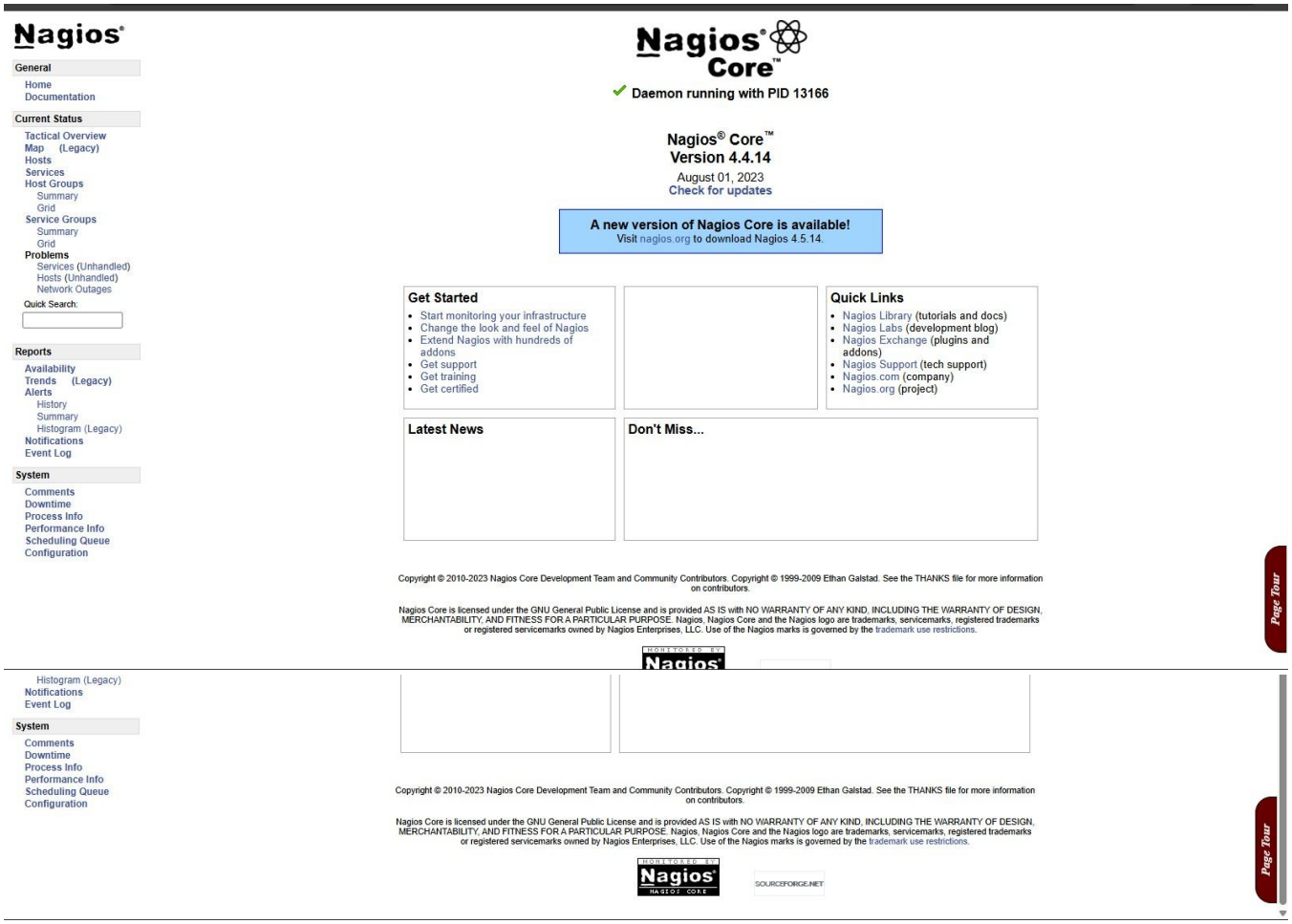
Remember, these are *SAMPLE* config files.  You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
        ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***
```

```
*** Nagios/Apache conf file installed ***

hp_15l@SExYSagar:/tmp/nagioscore-nagios-4.4.14$ sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
hp_15l@SExYSagar:/tmp/nagioscore-nagios-4.4.14$ sudo a2enmod cgi rewrite
sudo service apache2 restart
sudo service nagios start
Enabling module cgi.
Enabling module rewrite.
To activate the new configuration, you need to run:
  systemctl restart apache2
hp_15l@SExYSagar:/tmp/nagioscore-nagios-4.4.14$
```




Histogram (Legacy)
Notifications
Event Log
System
Comments
Downtime
Process Info
Performance Info
Scheduling Queue
Configuration
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Nagios Core is licensed under the GNU General Public License and is provided AS IS with NO WARRANTY OF ANY KIND, INCLUDING THE WARRANTY OF DESIGN, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE. Nagios, Nagios Core and the Nagios logo are trademarks, servicemarks, registered trademarks or registered servicemarks owned by Nagios Enterprises, LLC. Use of the Nagios marks is governed by the trademark use restrictions.
Page Tour

# Monitoring :

```
hp_15l@ubuntu:~$ sudo sed -i 's/Require valid-user/Require all granted/g' /etc/apache2/sites-enabled/nagios.conf
[sudo: authenticate] Password:
hp_15l@ubuntu:~$ sudo service apache2 restart
hp_15l@ubuntu:~$ sudo mkdir -p /usr/local/nagios/etc/servers
sudo tee /usr/local/nagios/etc/servers/MyWindowsHost001.cfg << 'EOF'
define host {
    use             windows-server
    host_name       Windows_Host_001
    alias           Windows Host 001
    address         127.0.0.1
    register        1
}

define service {
    host_name             Windows_Host_001
    service_description   PING
    check_command         check_ping!100.0,20%!500.0,60%
    max_check_attempts    2
    check_interval        2
    retry_interval        2
    check_period          24x7
    check_freshness       1
    contact_groups        admins
    notification_interval 2
    notification_period   24x7
    notifications_enabled 1
    register              1
}
EOF
define host {
    use             windows-server
    host_name       Windows_Host_001
    alias           Windows Host 001
    address         127.0.0.1
    register        1
}

define service {
    host_name             Windows_Host_001
    service_description   PING
    check_command         check_ping!100.0,20%!500.0,60%
    max_check_attempts    2
    check_interval        2
    retry_interval        2
```

```
hp_15l@ubuntu:~$ sudo sed -i 's|#cfg_dir=/usr/local/nagios/etc/servers|cfg_dir=/usr/
ocal/nagios/etc/servers|g' /usr/local/nagios/etc/nagios.cfg
hp_15l@ubuntu:~$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.c
g

Nagios Core 4.4.14
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2023-08-01
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
Error: Could not find any hostgroup matching 'windows-servers' (config file '/usr/lo
al/nagios/etc/servers/MyWindowsHost001.cfg', starting on line 1)
Error: Failed to process hostgroup names for host 'Windows_Host_001' (config file '/
sr/local/nagios/etc/servers/MyWindowsHost001.cfg', starting at line 1)
   Error processing object config files!


***> One or more problems was encountered while processing the config files...

     Check your configuration file(s) to ensure that they contain valid
     directives and data definitions.  If you are upgrading from a previous
     version of Nagios, you should be aware that some variables/definitions
     may have been removed or modified in this version.  Make sure to read
     the HTML documentation regarding the config files, as well as the
     'Whats New' section to find out what has changed.
```

```
hp_15l@ubuntu:~$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.4.14
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2023-08-01
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
Error: Could not find any hostgroup matching 'windows-servers' (config file '/usr/local/nagios/etc/servers/MyWindowsHost001.cfg', starting on line 1)
Error: Failed to process hostgroup names for host 'Windows_Host_001' (config file '/usr/local/nagios/etc/servers/MyWindowsHost001.cfg', starting at line 1)
   Error processing object config files!


***> One or more problems was encountered while processing the config files...

     Check your configuration file(s) to ensure that they contain valid
     directives and data definitions.  If you are upgrading from a previous
     version of Nagios, you should be aware that some variables/definitions
     may have been removed or modified in this version.  Make sure to read
     the HTML documentation regarding the config files, as well as the
     'Whats New' section to find out what has changed.

hp_15l@ubuntu:~$ sudo tee /usr/local/nagios/etc/servers/MyWindowsHost001.cfg << 'EOF'
define host {
    host_name           Windows_Host_001
    alias               Windows Host 001
    address             127.0.0.1
    max_check_attempts  3
    check_period        24x7
    notification_period 24x7
    notification_interval 30
    register            1
}

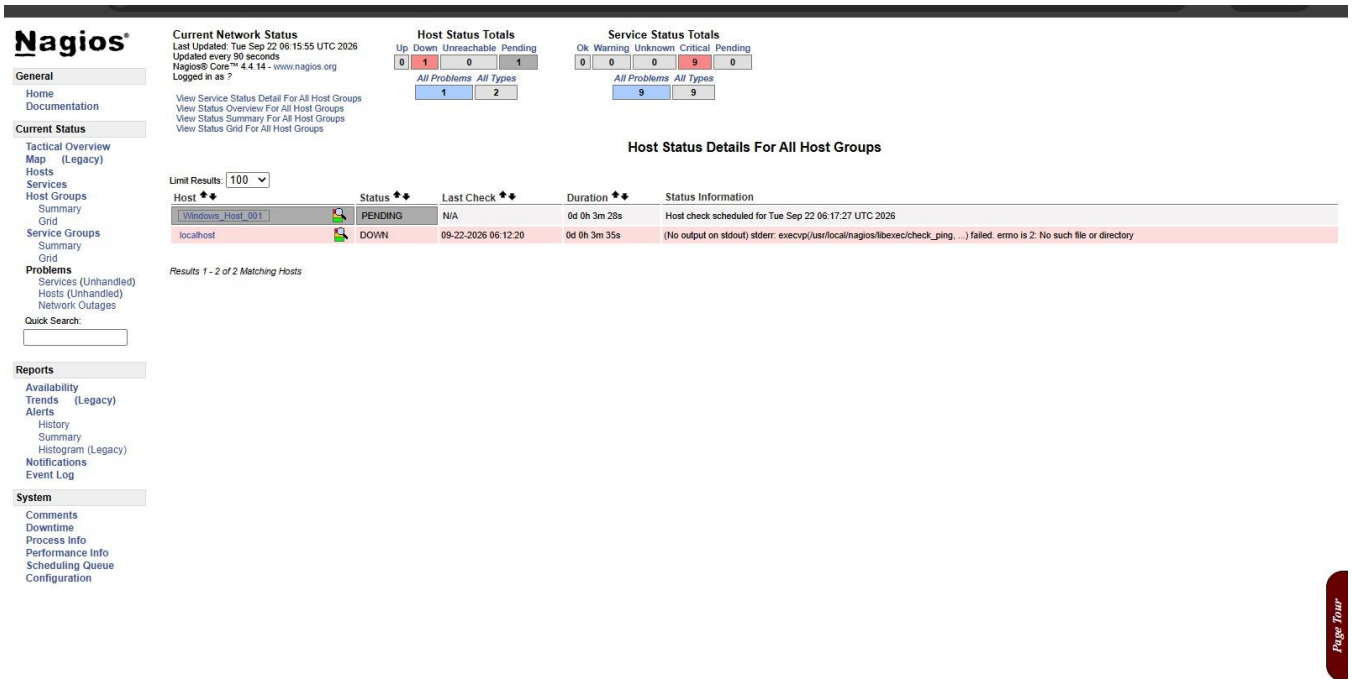
define service {
    host_name           Windows_Host_001
    service_description PING
    check_command       check_ping!100.0,20%!500.0,60%
    max_check_attempts  3
    check_interval      2
    retry_interval      1
```

```
hp_15l@ubuntu:~$ sudo tee /usr/local/nagios/etc/servers/MyWindowsHost001.cfg << 'EOF'
define host {
    host_name           Windows_Host_001
    alias               Windows Host 001
    address             127.0.0.1
    max_check_attempts  3
    check_period        24x7
    notification_period 24x7
    notification_interval 30
    register            1
}

define service {
    host_name           Windows_Host_001
    service_description PING
    check_command       check_ping!100.0,20%!500.0,60%
    max_check_attempts  3
    check_interval      2
    retry_interval      1
    check_period        24x7
    notification_period 24x7
    notification_interval 30
    register            1
}
EOF
define host {
    host_name           Windows_Host_001
    alias               Windows Host 001
    address             127.0.0.1
    max_check_attempts  3
    check_period        24x7
    notification_period 24x7
    notification_interval 30
    register            1
}

define service {
    host_name           Windows_Host_001
    service_description PING
    check_command       check_ping!100.0,20%!500.0,60%
    max_check_attempts  3
    check_interval      2
    retry_interval      1
```

```
efine service {
   host_name           Windows_Host_001
   service_description PING
   check_command       check_ping!100.0,20%!500.0,60%
   max_check_attempts  3
   check_interval      2
   retry_interval      1
   check_period        24x7
   notification_period 24x7
   notification_interval 30
   register            1
p_15l@ubuntu:~$ sudo service nagios start
p_15l@ubuntu:~$ sudo sed -i 's/use_authentication=1/use_authentication=0/g' /usr/local/nagios/etc/cgi.cfg
p_15l@ubuntu:~$ sudo service nagios restart
udo service apache2 restart
p_15l@ubuntu:~$
```




## Conclusion:

In this experiment, continuous IT infrastructure monitoring was performed using Nagios Core and remote monitoring agents. Nagios Core was configured with the Apache web server to provide a centralized monitoring dashboard. The NSClient++ agent was used to monitor remote Windows hosts and collect information about system performance and network availability through the NRPE protocol. Host and service checks, including ICMP ping tests, were configured to monitor connectivity and service status. The results were viewed and tracked through the Nagios web console. Overall, the experiment demonstrated how Nagios helps identify server issues early, monitor system performance, maintain service availability, and manage IT infrastructure efficiently.